

Verification Continuum™

FSDB Reader Summary

Version W-2024.09, September 2024



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Preface

The Verdi FSDB Reader Summary provides detailed information of the FSDB Reader updates which occurred in different Verdi versions.

Customer Support

For any online access to the self-help resources, you can refer to the documentation and searchable knowledge base available in SolvNetPlus.

To obtain support for your Verdi product, choose one of the following:

- Open a case through SolvNetPlus.

Go to <https://solvnetplus.synopsys.com/s/contactsupport> and provide the requested information, including:

- Product L1 as Verdi
- Case Type

Fill in the remaining fields according to your environment and issue.

- Send an e-mail message to verdi_support@synopsys.com.

Include product name (L1), sub-product name/technology (L2), and product version in your e-mail, so it can be routed correctly.

Your e-mail will be acknowledged by automatic reply and assigned a Case number along with Case reference ID in the subject (ref:____:ref).

For any further communication on this Case via e-mail, send e-mail to verdi_support@synopsys.com and ensure to have the same Case ref ID in the subject header or else it will open duplicate cases.

Note:

In general, we need to be able to reproduce the problem in order to fix it, so a simple model demonstrating the error is the most effective way for us to identify the bug. If that is not possible, then provide a detailed explanation of the problem along with complete error and corresponding code, if any/permissible.

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also

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includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

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Verdi FSDB Reader Summary

This chapter covers the previous release history for the FSDB reader.

Verdi 2023.12

Starting from this release, you can migrate to NPI Waveform Model/Writer, which is the latest FSDB Reader/Writer to seamlessly read FSDB. The following data types are supported only with NPI Waveform Model/Writer:

1. Modport/interface port
 2. Virtual file
 3. The FSDB which includes power signals
 4. Packed mode dumping
-

Verdi 2019.06

- Starting from this release, by default, Verdi filters the Value Change (VC) of Value Change Dump (VCD) real signal. To revert the previous behavior, that is, keeping two duplicates VC in real signal, set the environment variable `FSDB_KEEP_TWO_DUPLICATE_REAL_VC`.
 - Add `ffrGetAPIVersion` to get the FSDB Reader API version.
 - FSDB is upgraded from version 5.7 to version 5.8.
-

Verdi 2017.12

- Starting from this release, the FSDB is upgraded from version 5.5 to version 5.6.
- The `fsdbVarIdcode` is enlarged from `int` to `int64` so that larger designs (over 2G idcode) can be handled with ease in the future.

Verdi 2016.06

Starting from this release, functions and tasks can be dumped to FSDB. By default, they are not dumped for performance consideration. You can use one of following methods to dump them to FSDB during runtime (simulation)

1. Using the `FSDB_FUNCTIONS` environment variable:

For example: `setenv FSDB_FUNCTIONS 1`

2. Using the `+fsdb+functions` runtime option:

For example: `simv +fsdb+functions`

3. Adding the `+functions` option to your `$fsdbDumpvars` dumping task:

For example: `$fsdbDumpvars "+functions")`

4. Adding the `+functions` option to your `fsdbDumpvars` Tcl command:

For example: `fsdbDumpvars +functions`

Verdi 2015.09

Starting from this release, the support for FSDB API static library(.a) is stopped.

Verdi 2014.12

Add `ffrGetVersion` to get the FSDB library version.

Verdi 2014.03

Add `ffrOpenNonSharedObj` to return a different address of the reader object if an FSDB file is opened multiple times in the same job.

Verdi 2013.07

Add `ffrFreeNavDB` to free all NavDB that are used by signals in the signal list.

Verdi 2013.04

Add `ffrNoChangeEpicVarName` to prevent the slash (/) character in the NanoSim FSDB (type `FSDB_FT_NANOSIM`) signal name from being converted to the caret (^) character.

Verdi 2012.07

- Add `ffrIsVrtlFile` to check whether the specified file is a virtual file and `ffrGetVrtlFileType` to get its file type.
- Add `ffrGetFileDumperType` to get the dumper type.
- Add `ffrGetFileSimType` to get the simulator type.

Verdi 2012.01

The Verdi version includes a newly-engineered FSDB that has been optimized for significant size savings. The version of this new FSDB file format is FSDB 5.0.

FSDB files generated by the Verdi 2012.01 FSDB dumper or writer can only be loaded into Verdi 2012.01 or later versions.

- The FSDB 5.0 version for the Verdi Reader has exactly the same API interface as the FSDB 4.x (Novas 2011.10) version. Header files are not changed.
- The Verdi Reader can read the FSDB 5.0 format and earlier formats.
- For FSDB 5.0 files, the Verdi Reader automatically uses a parallel scheme to leverage multi-threaded reading and improves the FSDB loading speed. For the FSDB 4.3 version or any earlier version files, the Verdi Reader can read these files, but the parallel scheme will not be used. To turn the parallel scheme off, use the `FFR_MT_THREAD_COUNT` environment variable:

```
setenv FFR_MT_THREAD_COUNT n
```

When $n > 0$, the specified number of threads is used. If n is bigger than the machine core number, n is adjusted to the core number automatically.

When $n = 1$, the parallel scheme is turned off.

- **Note:**
 1. Multi-threaded reading is supported only on LINUX platforms.
 2. In multi-threaded reading, multiple threads usually consume more memory. Each additional FSDB reader thread consumes approximately 10 Mega-Bytes (MB) of memory.